TEACHER FOCUS GUIDE

FTC StarterBot TeleOp Lesson Focus Guide

Part 8 Video Pauses and Classroom Facilitation

Recommended use | Video discussion plus a controlled hands-on test

Use this guide beside the Part 8 video to keep discussion focused on the relationship among code, configuration, controls, and physical motion. The final goal is not simply to make the robot move; students should be able to predict each command and defend a diagnosis with evidence.

Before students arrive

- [ ] Secure the robot on stable blocks with wheels and intake clear.
- [ ] Verify the corrected Java program and all five configured names.
- [ ] Display the POWERING ON poster and mark the test boundary.
- [ ] Open the video at https://youtu.be/5OZR2QvJUuE.
- [ ] Print one student workbook per student or pair.

Core vocabulary

iterative OpMode | hardwareMap | configuration name | split arcade | forward | rotation | normalize | CRServo | trigger subtraction | telemetry

Lesson success criteria

- Students calculate outputs before testing.
- Students keep code variables separate from configuration names.
- Students explain the purpose of motor and servo reversal.
- Students follow the full call-and-clear sequence without prompting.
- Students compare prediction, telemetry, and movement before changing code.

Teacher pause questions

PAUSE	ASK	LISTEN FOR
0:09	What are the three robot jobs controlled by one gamepad?	Drive the chassis, run the intake roller, and move both corner feeders.
0:19	Which OpMode methods run once, and which method repeats?	init and stop run once; loop repeats after START.

PAUSE	ASK	LISTEN FOR
0:31	Why must each quoted hardware name match exactly?	hardwareMap uses the quoted string to locate the configured device.
0:42	Why is one drive motor reversed, and what does BRAKE actually do?	Mirrored motors need opposite code directions to agree on forward. BRAKE reduces coasting; it is not a mechanical lock.
0:53	What must happen after the POWERING ON call and before START?	Everyone steps away, the operator pauses, and the safety lead visually confirms the entire area is clear.
1:14	Which controls provide forward and rotation?	The left stick Y controls forward and backward; the right stick X controls rotation.
1:26	Why does one drive equation add rotation while the other subtracts it?	Different left and right powers create the speed difference needed to turn.
1:39	What happens when both intake triggers are pressed equally?	Their values subtract to zero, so the intake command is zero.
1:48	How does one intakePower value coordinate three devices?	The same value is sent to the roller and both feeder servos; one mirrored servo is reversed.
1:56	How can telemetry help separate a code problem from a hardware problem?	It shows calculated commands and inputs so students can compare expected values with physical behavior.

Fast hands on sequence

STEP	ACTION	STUDENT EVIDENCE
1	Map	Match five Java variables, quoted names, and physical devices while disabled.
2	Predict	Calculate one straight, one turning, and one curved arcade-drive case.
3	Calculate	Use right trigger minus left trigger for inward, outward, and stopped intake cases.
4	Clear	Run the complete POWERING ON sequence and wait for visual confirmation.

STEP	ACTION	STUDENT EVIDENCE
5	Test	Use small inputs with the robot raised. Return controls to neutral after each test.
6	Verify	Compare the code calculation, telemetry value, and observed motion.
7	Stop	Press STOP and confirm DISABLED before anyone crosses the boundary.

Common misconceptions

MISCONCEPTION	CORRECTION
The Java variable must have the same spelling as the configured name	Only the quoted hardwareMap string must match the Robot Controller configuration exactly.
BRAKE locks the robot	BRAKE reduces coasting after zero power; it does not create a safe mechanical lock.
CRServo setPower selects a position	For a continuous-rotation servo, power controls speed and direction.
Telemetry proves the motor moved	Telemetry shows the program's data. Physical observation verifies hardware response.
A stopped wheel means the robot is safe to approach	The team must press STOP and visually confirm DISABLED before approach.

Teacher closing prompt. Ask each student to name one command path from gamepad input to physical device and one safety or diagnostic check that belongs in that path.